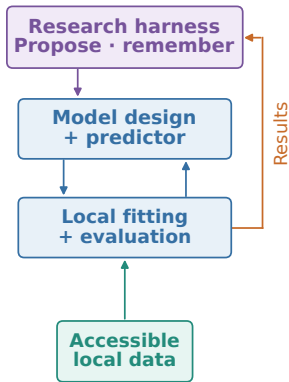


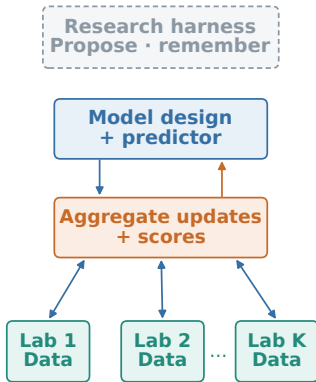
(a) Centralized bio-agent

Centralized access



(b) Collaborative training

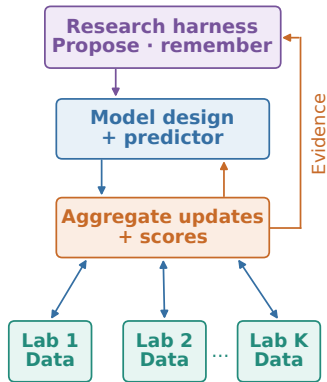
Fixed design



Train locally; share updates

(c) BioCoLoop

Model + design learning



Train locally; share updates

□ Data

□ Predictor

□ Research loop

□ Aggregation

Dashed gray: inactive harness. K denotes laboratories.